

Homework 4

Out: April 6th, 2026, Monday -- Due: April 22nd, 2026, Wednesday, 11:59pm

EC327 Introduction to Software Engineering – Spring 2026

Total: 50 points

Submission

- ✓ Create 1 PDF for all questions combined
- ✓ There will be only 1 Gradescope submission which contains all questions
- ✓ Submit via Gradescope

Q1. [30 points] – PA3 Virtual Tables

Create a diagram for the virtual table relationships in PA3.

1. Determine which classes have virtual tables
2. For each class that has a virtual table, create a UML entity for that class
 - a. In particular, focus on the functions only in the UML diagram. Member variables are not needed.
3. Then for each of those classes with virtual tables, create a table for the virtual table itself.
 - a. Include a pointer in the class to the virtual table in the UML diagram
4. Draw arrow(s) that show the mapping between the functions in the virtual table and the functions that will be called in the UML diagram.

Feel free to use the internet for example diagrams, styles, etc.

Q2. [20 points] – PA3 State Diagram

Create a state diagram for the Mage behavior in PA3. A state diagram consists of circles for states (e.g. stopped, moving, etc.) and labeled arcs between these states. The labels should indicate what action (i.e. function call or variable values) causes the Mage to change state. Also label the states (circles) with the state character the Mage will have in those states. Be sure to include “self-loops” for states that the Trainer can remain in.